

4.12.13

Chiruvella Harshith Sharan - AI25BTECH11008

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Question

If the equation of the base of an equilateral triangle is

$$x + y = 2, \quad \text{and the vertex is } (2, -1),$$

find the length of the side of the triangle.

Step 1: Vector Form

Equation of base line:

$$\vec{n}^T \vec{x} = 2 \quad (1)$$

where $\vec{n} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Vertex:

$$\vec{A} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

Step 2: Foot of Perpendicular

Distance from $\vec{A}$ to line:

$$d = \frac{\vec{n}^T \vec{A} - 2}{\|\vec{n}\|} = \frac{-1}{\sqrt{2}} \quad (2)$$

Foot of perpendicular:

$$\vec{D} = \vec{A} - d \frac{\vec{n}}{\|\vec{n}\|} = \begin{pmatrix} \frac{5}{2} \\ -\frac{1}{2} \end{pmatrix}.$$

Step 3: Altitude Relation

In an equilateral triangle:

$$AD = \frac{\sqrt{3}}{2}a \quad (3)$$

Here

$$AD = \|\vec{A} - \vec{D}\| = \frac{1}{\sqrt{2}}.$$

Step 4: Side Length

Using (3):

$$\frac{1}{\sqrt{2}} = \frac{\sqrt{3}}{2}a \implies a = \sqrt{\frac{2}{3}} \quad (4)$$

$$\boxed{a = \sqrt{\frac{2}{3}}}$$

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

# Given points
A = np.array([2, -1])      # Vertex
D = np.array([2.5, -0.5]) # Foot of perpendicular

# Side length of equilateral triangle
a = np.sqrt(2/3)

# Direction vector of the base (perpendicular to
# normal (1,1))
m = np.array([1, -1]) / np.sqrt(2) # Unit vector
# along base

# Base vertices B and C
B = D + (a/2) * m
C = D - (a/2) * m
```

Python Code

```
# Line x+y=2
x = np.linspace(-2, 4, 400)
y = 2 - x

# Plot base line
plt.plot(x, y, 'k-', label=r"$x+y=2$")

# Plot triangle edges
plt.plot([A[0], B[0]], [A[1], B[1]], 'b-')
plt.plot([A[0], C[0]], [A[1], C[1]], 'b-')
plt.plot([B[0], C[0]], [B[1], C[1]], 'b-')

# Plot altitude AD
plt.plot([A[0], D[0]], [A[1], D[1]], 'r--', label="
    Altitude")
```


Python Code

```
# Mark points
plt.scatter(*A, color='blue')
plt.text(A[0]+0.1, A[1]-0.2, "A(2,-1)", fontsize=10)

plt.scatter(*D, color='green')
plt.text(D[0]+0.1, D[1], "D(2.5,-0.5)", fontsize=10)

plt.scatter(*B, color='purple')
plt.text(B[0]+0.1, B[1], "B", fontsize=10)

plt.scatter(*C, color='purple')
plt.text(C[0]+0.1, C[1], "C", fontsize=10)
```

Python Code

```
# Axes settings
plt.axhline(0, color='black', linewidth=0.5)
plt.axvline(0, color='black', linewidth=0.5)
plt.gca().set_aspect('equal', adjustable='box')
plt.grid(True, linestyle='--', alpha=0.6)
plt.legend()
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("Equilateral Triangle with base  $x+y=2$  and
          vertex A(2,-1)")

plt.show()
```

C Code

```
#include <stdio.h>
#include <math.h>

int main() {
    double Ax=2, Ay=-1;
    double Dx=2.5, Dy=-0.5;

    double AD = sqrt((Ax-Dx)*(Ax-Dx) + (Ay-Dy)*(Ay-Dy)
        );
    double a = (2 * AD) / sqrt(3);

    printf("Side length = %lf\n", a);
    return 0;
}
```

Plot

beamer/figs/fig1.png